THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8	$(m+n)^2 - m^2$
2	**(a)** 6	Find b by Pythagoras
3	**(d)** $\frac{12}{5}$	$\frac{1}{2}ab = \frac{1}{2}ch$
4	**(b)** 14	Sides in ratio $a : b$
5	**(c)** $\frac{9}{5}$	$AB^2 = AD \cdot AC$
6	**(d)** 25	Side $= s$
7	**(b)** 21	$(m+n)^2 - m^2$
8	**(b)** 30	Find b by Pythagoras
9	**(a)** $\frac{60}{13}$	$\frac{1}{2}ab = \frac{1}{2}ch$
10	**(c)** 18	Sides in ratio $a : b$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{7}$ (1); $DE = 6$ cm (1).	1+1+1
12	Ratio of areas $=$ (ratio of sides)^2 (1); $\frac{27}{75} = \frac{3^2}{x^2}$ (1); $x = 5$ cm (1).	1+1+1
13	Hypotenuse 5 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{12}{5}$ cm (1).	1+1
14	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{2}{7}$ (1); $DE = 6$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 2:3$ (1); area ratio $= 4:9$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (6)^2 - 3^2 = 27$ (1+1); $h = 3\sqrt{3}$ cm (1).	1+1+1
17	$AC = 5$ cm (1); $AD = \frac{AB^2}{AC} = \frac{9}{5}$ cm, $DC = \frac{BC^2}{AC} = \frac{16}{5}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{12}{5}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 3:5$ (1); area ratio $= 9:25$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (8)^2 - 4^2 = 48$ (1+1); $h = 4\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 13$ cm (1); $AD = \frac{AB^2}{AC} = \frac{25}{13}$ cm, $DC = \frac{BC^2}{AC} = \frac{144}{13}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{60}{13}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Right triangle with legs 60 and 80 (1); distance $= \sqrt{3600 + 6400} = 100$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Draw a horizontal line from the top of the shorter pole: right triangle with legs 24 and 18 (2); $x^2 = 576 + 324$ (1); $x = 30$ m (2).	2+1+2
20	Right triangle with legs 50 and 120 (1); distance $= \sqrt{2500 + 14400} = 130$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Draw a horizontal line from the top of the shorter pole: right triangle with legs 27 and 36 (2); $x^2 = 729 + 1296$ (1); $x = 45$ m (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{25}{13}$	$AB^2 = AD \cdot AC$
2	**(c)** 49	Side $= s$
3	**(a)** 40	$(m+n)^2 - m^2$
4	**(b)** 60	Find b by Pythagoras
5	**(b)** $\frac{120}{17}$	$\frac{1}{2}ab = \frac{1}{2}ch$
6	**(c)** 22	Sides in ratio $a : b$
7	**(a)** $\frac{64}{17}$	$AB^2 = AD \cdot AC$
8	**(c)** 81	Side $= s$
9	**(b)** 24	$(m+n)^2 - m^2$
10	**(c)** 84	Find b by Pythagoras

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ratio of areas = (ratio of sides)^2 (1); $\frac{32}{162} = \frac{4^2}{x^2}$ (1); $x = 9$ cm (1).	1+1+1
12	Hypotenuse 13 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{60}{13}$ cm (1).	1+1
13	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{4}{9}$ (1); $DE = 8$ cm (1).	1+1+1
14	Ratio of areas = (ratio of sides)^2 (1); $\frac{20}{245} = \frac{2^2}{x^2}$ (1); $x = 7$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 1:4$ (1); area ratio $= 1:16$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (10)^2 - 5^2 = 75$ (1+1); $h = 5\sqrt{3}$ cm (1).	1+1+1
17	$AC = 17$ cm (1); $AD = \frac{AB^2}{AC} = \frac{64}{17}$ cm, $DC = \frac{BC^2}{AC} = \frac{225}{17}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{120}{17}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 2:7$ (1); area ratio $= 4:49$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (12)^2 - 6^2 = 108$ (1+1); $h = 6\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 25$ cm (1); $AD = \frac{AB^2}{AC} = \frac{49}{25}$ cm, $DC = \frac{BC^2}{AC} = \frac{576}{25}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{168}{25}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Right triangle with legs 48 and 90 (1); distance $= \sqrt{2304 + 8100} = 102$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Distance walked $= 8$ m; let shadow $= x$ m (1); similar triangles: $\frac{x}{x+8} = \frac{2}{6}$ (2); $x = 4$ m (2).	1+2+2
20	Right triangle with legs 90 and 120 (1); distance $= \sqrt{8100 + 14400} = 150$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Distance walked $= 12$ m; let shadow $= x$ m (1); similar triangles: $\frac{x}{x+12} = \frac{3}{9}$ (2); $x = 6$ m (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Triangles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{168}{25}$	$\frac{1}{2}ab = \frac{1}{2}ch$
2	**(d)** 16	Sides in ratio $a : b$
3	**(c)** $\frac{18}{5}$	$AB^2 = AD \cdot AC$
4	**(a)** 121	Side $= s$
5	**(a)** 45	$(m+n)^2 - m^2$
6	**(a)** 210	Find b by Pythagoras
7	**(c)** $\frac{420}{29}$	$\frac{1}{2}ab = \frac{1}{2}ch$
8	**(d)** 20	Sides in ratio $a : b$
9	**(a)** $\frac{27}{5}$	$AB^2 = AD \cdot AC$
10	**(d)** 169	Side $= s$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Hypotenuse 17 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{120}{17}$ cm (1).	1+1
12	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{5}{8}$ (1); $DE = 20$ cm (1).	1+1+1
13	Ratio of areas = (ratio of sides)^2 (1); $\frac{50}{72} = \frac{5^2}{x^2}$ (1); $x = 6$ cm (1).	1+1+1
14	Hypotenuse 25 cm (1); $\frac{1}{2}ab = \frac{1}{2}ch$ so $h = \frac{168}{25}$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 3 : 4$ (1); area ratio $= 9 : 16$ (1).	1+1+1
16	Altitude bisects the base: $h^2 = (14)^2 - 7^2 = 147$ (1+1); $h = 7\sqrt{3}$ cm (1).	1+1+1
17	$AC = 29$ cm (1); $AD = \frac{AB^2}{AC} = \frac{400}{29}$ cm, $DC = \frac{BC^2}{AC} = \frac{441}{29}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{420}{29}$ cm (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\frac{AO}{OC} = \frac{AB}{DC} = 4 : 9$ (1); area ratio $= 16 : 81$ (1).	1+1+1
18	Altitude bisects the base: $h^2 = (18)^2 - 9^2 = 243$ (1+1); $h = 9\sqrt{3}$ cm (1).	1+1+1
	OR: $AC = 41$ cm (1); $AD = \frac{AB^2}{AC} = \frac{81}{41}$ cm, $DC = \frac{BC^2}{AC} = \frac{1600}{41}$ cm (1); $BD = \frac{AB \cdot BC}{AC} = \frac{360}{41}$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Draw a horizontal line from the top of the shorter pole: right triangle with legs 12 and 16 (2); $x^2 = 144 + 256$ (1); $x = 20$ m (2).	2+1+2
	OR: Distance walked $= 20$ m; let shadow $= x$ m (1); similar triangles: $\frac{x}{x+20} = \frac{1}{5}$ (2); $x = 5$ m (2).	1+2+2
20	Draw a horizontal line from the top of the shorter pole: right triangle with legs 9 and 12 (2); $x^2 = 81 + 144$ (1); $x = 15$ m (2).	2+1+2
	OR: Distance walked $= 12$ m; let shadow $= x$ m (1); similar triangles: $\frac{x}{x+12} = \frac{2}{8}$ (2); $x = 4$ m (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.